

RAMALINGAM T

ramalingamthangamani2023@gmail.com | +91 6374517342 | Erode

PROFILE SUMMARY

AI & Data Science undergraduate with research and industry internship experience at IITM Pravartak and IGRAD AI Labs. Passionate about deploying intelligent systems that solve real-world problems, with hands-on experience in machine learning, NLP, and computer vision – from edge inference on wearables to offline voice assistants and video pipelines. Eager to contribute to AI-driven products in automation and smart systems.

EDUCATION

K S Rangasamy College of Technology, Tiruchengode 2023 – 2027

- B.Tech Artificial Intelligence and Data Science

CGPA – 7.7

SKILLS

- AI / Machine Learning Technologies:** Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Transformers, Multimodal AI Systems, Real-Time AI Inference, Edge AI Integration, Model Training & Deployment, Model Optimization & Hyperparameter Tuning, Data Preprocessing, MLOps, DevOps
- Programming:** Python, Java, SQL
- Frameworks & Libraries:** OpenCV, MoviePy, Whisper, SpeechBrain
- Tools & Others:** GitHub, Google Colab, n8n, Kaggle, Roboflow
- 3D & XR Development:** Unity, Blender, VR/XR Development
- UI/UX Design:** Figma, Adobe Illustrator, Wireframing, Prototyping, Design Systems

PORTFOLIO & PROFILES

- LinkedIn:** <https://www.linkedin.com/in/ramalingam05>
- GitHub:** <https://github.com/ramalingamthangamani>
- Portfolio:** <https://ramalingamthangamani.github.io/Portfolio>

CERTIFICATIONS

- Virtual Internship – AI/ML Essentials – AICTE, 2024
- Generative AI Course Completion – Google Learn Platform, 2025
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate – Oracle, 2025
- Elements of AI – University of Helsinki, 2026
- Natural Language Processing Course – Hugging Face, 2026

INTERNSHIP EXPERIENCE

IITM Pravartak Technologies Foundation | Mar 2026 – Sep 2026 | AI Research Intern

- Researched and developed AI-powered solutions using machine learning, deep learning, and generative AI. Contributed to model experimentation, data preprocessing, performance evaluation, and prototype development alongside research teams working on real-world engineering problems.

IGRAD AI Labs Pvt. Ltd. | Sep 2025 – Nov 2025 | AI Developer Intern

- Developed an AI-powered system for automatic cricket highlight generation using computer vision and deep learning. Implemented end-to-end video clipping and merging pipelines with Python, OpenCV, and MoviePy to streamline highlight creation.

PROJECTS

IWI – Intelligent Wearable Interface

- Developed an AI-powered wearable glove that performs real-time gesture recognition, enabling gesture-to-text/speech translation, human–computer interaction, and assistive communication through low-latency edge ML inference.

Thozhan – Offline Desktop Intelligence

- Built a secure, fully offline AI voice assistant with speaker authentication, Tanglish speech understanding, and intent-based command execution. Integrated Whisper for speech recognition, SpeechBrain for voice verification, and custom NLP models to enable real-time, voice-controlled system automation.

AI Prescription Parser – Automated Registration Suite

- Built an OCR + NLP system that transforms handwritten and digital prescriptions into structured patient records. Automates patient registration, medical form filling, data validation, and electronic record creation, reducing manual processing and improving healthcare workflow efficiency.

PortInspector – Cross-Platform Port Monitoring CLI

- Developed a cross-platform command-line tool for monitoring and analyzing network ports in real time. Enables users to scan open ports, identify running services, inspect port usage, and quickly diagnose network connectivity and process-level issues through a lightweight and developer-friendly CLI.

Agentic System Analyzer & Optimizer

- Designed an intelligent platform for evaluating AI applications by analyzing prompts, models, APIs, and workflow pipelines. Detects performance bottlenecks, measures efficiency, and recommends optimizations to improve accuracy, scalability, response time, and cost-effectiveness.

Social Radar – Real-Time Trend Analysis

- Built a real-time social intelligence platform that aggregates data from multiple social media platforms worldwide. Enables hashtag and keyword-based trend discovery, sentiment analysis, engagement tracking, and visualization of emerging global conversations through AI-driven analytics.

ACTIVITIES

- Attended Supercomputing India 2025 at MIT, Bangalore.
- Conducted hands-on workshops on AI for Game Development at KSRTC.
- Presented at a Technical Symposium at Karpagam College of Engineering.